Nonlinear equations in one variable and systems of equations in two variables — Part 1

Advanced Math · Math · 82 questions

1.1 a87d4ca6-6608-46d3-9f6b-60f4ad868594 · Easy

$$x = 49$$

$$y = \sqrt{x} + 9$$

The graphs of the given equations intersect at the point x, y in the xy-plane. What is the value of y?

A. 16

B. 40

C. 81

D. 130

1.2 04095-DC · Easy

$6x - 9y > 12$

Which of the following inequalities is equivalent to the inequality above?

A. $x - y > 2$

B. $2x - 3y > 4$

C. $3x - 2y > 4$

D. $3y - 2x > 2$

1.3 7fb0079b-bb84-4295-8b22-1174f114e4e5 · Easy

$$p + 61 = 65$$

Which value is a solution to the given equation?

A. $\frac{65}{61}$

B. 4

C. 126

D. 130

1.4 08137-DC · Easy

Which of the following is a solution to the equation $2x^2 - 4 = x^2$?

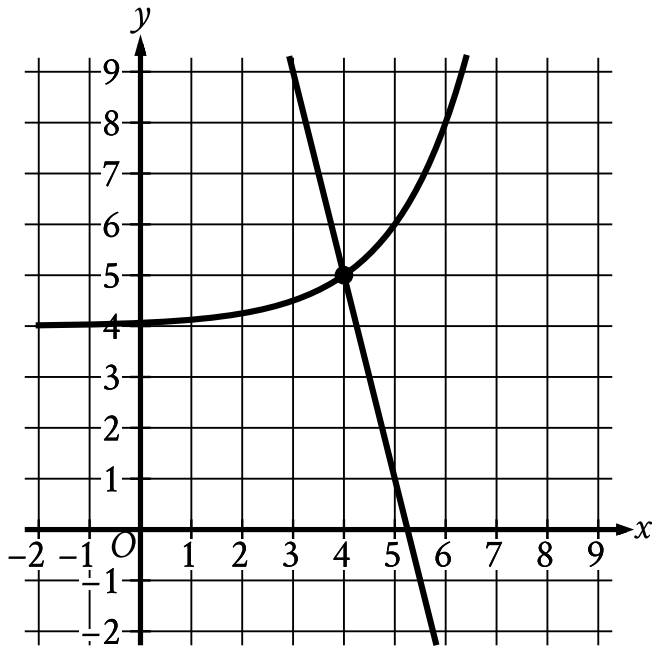
A. 1

B. 2

C. 3

D. 4

1.5 39a54602-89bd-4494-9358-17d9877e73da · Easy



- For the curve in the system:
  - The curve passes from quadrant 2 to quadrant 1.
  - In quadrant 2, the curve trends up gradually to the approximate point (0 comma 4.063).
  - In quadrant 1:
    - The curve trends up gradually to point (4 comma 5).
    - The curve then trends up sharply.
  - The curve passes through the following points:
    - (3 comma nine halves)
    - (4 comma 5)
    - (5 comma 6)
- For the line in the system:
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (3 comma 9)
    - (4 comma 5)

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution x, y to this system?

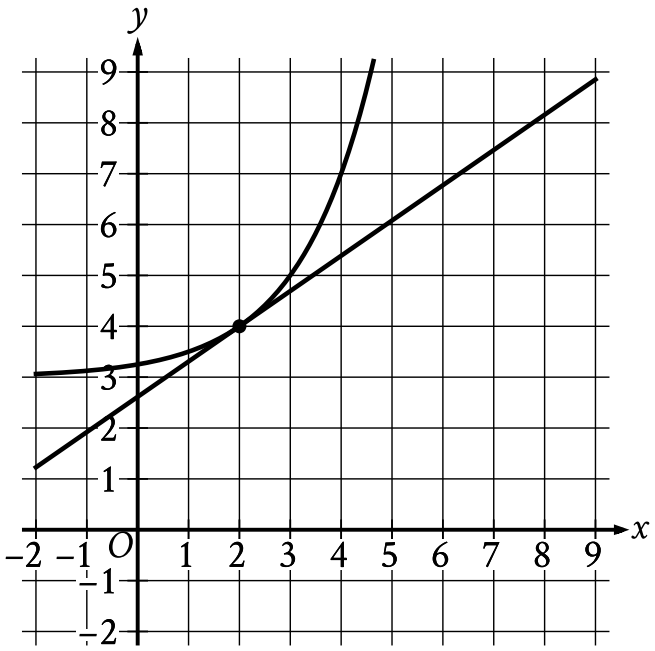
A. $0, 0$

B. $0, 4$

C. $4, 5$

D. $5, 0$

1.6 e918de33-af58-440c-b5a8-b50469fe0230 · Easy



- For the curve in the system:
  - Moving from left to right:
    - The curve passes from quadrant 2 to quadrant 1.
    - In quadrant 2, the curve trends up gradually to point (0 comma 3.25).
    - In quadrant 1, the curve trends up sharply.
  - The curve passes through the following points:
    - (1 comma 3.5)
    - (2 comma 4)
    - (4 comma 7)
- For the line in the system:
  - The line slants gradually up from left to right.
  - The line passes through the following points:
    - (2 comma 4)
    - (7 comma 7.5)

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution x, y to this system?

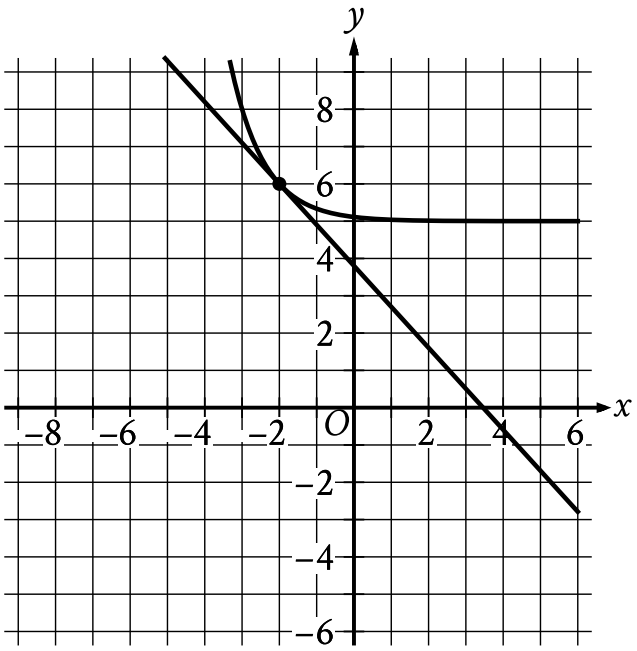
A. $0, 0$

B. $0, 2$

C. $2, 4$

D. $4, 0$

1.7 6a66013a-b796-4e4b-addb-dcd19f49d6f3 · Easy



- For the line in the system:
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (negative 2 comma 6)
    - approximately (0 comma 3.8)
- For the curve in the system:
  - Moving from left to right:
    - The curve passes from quadrant 2 to quadrant 1.
    - In quadrant 2:
      - The curve trends down sharply to the point (negative 2 comma 6).
      - The curve then trends down gradually to the y axis.
    - In quadrant 1, the curve trends down gradually.
  - As x increases, the curve approaches the line y equals 5.
  - The curve passes through the following points:
    - (negative 3 comma 8)
    - (negative 2 comma 6)

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution x, y to this system?

A. $6, 0$

B. $-2, 6$

C. $0, -2$

D. $0, 0$

1.8 31e60f67-d044-449f-abb3-8104b7c7e268 · Easy

$$7m = 2n + p$$

The given equation relates the positive numbers m, n, and p. Which equation correctly gives m in terms of n and p?

A. $m = \frac{2n + p}{7}$

B. $m = 2n + p$

C. $m = 2n + p - 7$

D. $m = 2 - n - p - 7$

1.9 bb261990-bccf-4eea-bddb-5b86e004de1f · Easy

$$x - 5 = 10$$

What is one possible solution to the given equation?

Student-produced response: _____

1.10 711ce341-6b08-4443-a2bc-e3ddb891f609 · Easy

$$5x = 45$$

What is the positive solution to the given equation?

Student-produced response: _____

1.11 7ba8f490-3a61-4ed0-b1ca-fedbddefc140 · Easy

$$\frac{r}{38} = s + t$$

The given equation relates the quantities r, s, and t. Which equation correctly expresses r in terms of s and t?

A. $r = s + t + 38$

B. $r = s + t - 38$

C. $r = 38s + t$

D. $r = \frac{s+t}{38}$

1.12 35b766b9-da19-4203-90cb-2032ce081aac · Easy

$$q - 29r = s$$

The given equation relates the positive numbers q, r, and s. Which equation correctly expresses q in terms of r and s?

A. $q = s - 29r$

B. $q = s + 29r$

C. $q = 29rs$

D. $q = -\frac{s}{29r}$

1.13 5014724c-eaa2-486b-8c20-ebe36ecc982b · Easy

$$y = 76$$
$$y = x^2 - 5$$

The graphs of the given equations in the xy-plane intersect at the point (x, y). What is a possible value of x?

A. $-\frac{76}{5}$

B. -9

C. 5

D. 76

1.14 016505-DC · Easy

The total revenue from sales of a product can be calculated using the formula $T = PQ$, where T is the total revenue, P is the price of the product, and Q is the quantity of the product sold. Which of the following equations gives the quantity of product sold in terms of P and T ?

A. $Q = \frac{P}{T}$

B. $Q = \frac{T}{P}$

C. $Q = PT$

D. $Q = T - P$

1.15 43b67788-a46c-45bd-a2de-2d26a4afa7b3 · Easy

$$3x^2 - 15x + 18 = 0$$

How many distinct real solutions are there to the given equation?

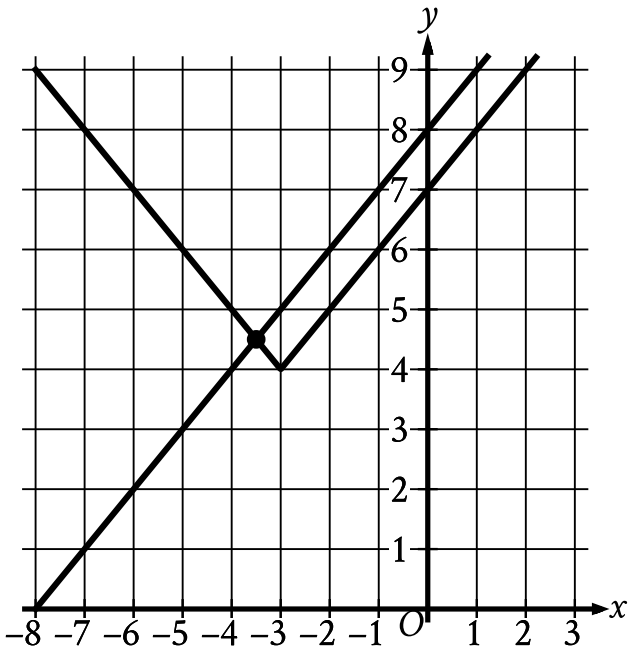
A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

1.16 dad4b1da-ac0d-4094-a181-0c9e028aef62 · Easy



- For the absolute value function in the system:
  - Moving from left to right:
    - The function slants sharply down to the point (negative 3 comma 4).
    - The function then slants sharply up.
  - The function passes through the following points:
    - (negative seven halves comma nine halves)
    - (negative 3 comma 4)
    - (negative five halves comma nine halves)
- For the linear function in the system:
  - The function slants sharply up from left to right.
  - The function passes through the following points:
    - (negative seven halves comma nine halves)
    - (0 comma 8)

The graph of a system of an absolute value function and a linear function is shown. What is the solution x, y to this system of two equations?

A. $0, 8$

B. $\frac{7}{2}, \frac{9}{2}$

C. $-\frac{7}{2}, \frac{9}{2}$

D. $-3, 4$

1.17 032680-DC · Easy

$$x+y=12$$
$$y=x^2$$

If (x,y) is a solution to the system of equations above, which of the following is a possible value of x?

A. 0

B. 1

C. 2

D. 3

1.18 022943-DC · Easy

If $(x+5)^2=4$, which of the following is a possible value of x ?

A. 1

B. -1

C. -2

D. -3

1.19 0dbf87b9-1746-41f3-8964-f290345cd695 · Easy

$$k^2 - 53 = 91$$

What is the positive solution to the given equation?

A. 144

B. 72

C. 38

D. 12

1.20 1b4d39f7-ca42-438f-8931-7679b4d7cc6d · Easy

$$x + 2x - 5x + 9 = 0$$

What is a positive solution to the given equation?

A. 3

B. 4

C. 5

D. 18

1.21 8fd03417-fa6c-4681-901b-22278562189d · Easy

$$y = 5x + 4$$

$$y = 5x^2 + 4$$

Which ordered pair x, y is a solution to the given system of equations?

A. $0, 0$

B. $0, 4$

C. $8, 44$

D. $8, 84$

1.22 cd40484a-aae7-48a6-883c-e658d661353a · Easy

$$x + 7 = 10$$

$$x + 7^2 = y$$

Which ordered pair x, y is a solution to the given system of equations?

A. $3, 100$

B. $3, 3$

C. $3, 10$

D. $3, 70$

1.23 d2ce343c-ce62-40d3-805d-1ec24af622ab · Easy

$$6r = 7s + t$$

The given equation relates the variables r, s, and t. Which equation correctly expresses s in terms of r and t?

A. $s = 42r - t$

B. $s = 76r - t$

C. $s = \frac{6}{7}r - t$

D. $s = \frac{6r - t}{7}$

1.24 29c9eada-06c0-4ac4-846d-366083f6cf03 · Easy

$$2r = 9s + t$$

The given equation relates the variables r, s, and t. Which equation correctly expresses s in terms of r and t?

A. $s = \frac{2}{9}r - t$

B. $s = \frac{2r - t}{9}$

C. $s = 18r - t$

D. $s = 92r - t$

1.25 6130d2b7-20eb-4881-8cfa-8ee3adb6499e · Easy

$$8j = k + 15m$$

The given equation relates the distinct positive numbers j, k, and m. Which equation correctly expresses j in terms of k and m?

A. $j = \frac{k}{8} + 15m$

B. $j = k + \frac{15m}{8}$

C. $j = 8(k + 15m)$

D. $j = \frac{k + 15m}{8}$

1.26 019519-DC · Easy

$$P = \frac{W}{t}$$

The power P produced by a machine is represented by the equation above, where W is the work performed during an amount of time t. Which of the following correctly expresses W in terms of P and t ?

A. $W = Pt$

B. $W = \frac{P}{t}$

C. $W = \frac{t}{P}$

D. $W = P + t$

1.27 c0817724-5c40-441b-9e3b-7d5c5dc9856c · Easy

$$c - 7 = 25p + k$$

The given equation relates the positive numbers c, p, and k. Which equation correctly expresses c in terms of p and k?

A. $c = 25p + k + 7$

B. $c = 25p + k - 7$

C. $c = 725p + k$

D. $c = \frac{25p + k}{7}$

1.28 67266c3c-00b9-45bd-91e8-fbde7f826062 · Easy

$$\frac{1}{x} + 8 = \frac{1}{6} + 8$$

What is the solution to the given equation?

A. 1

B. 6

C. 8

D. 14

1.29 108c96b1-9f6f-4078-9044-a3fccd85c37c · Easy

$$x + 45 = 48$$

What is the positive solution to the given equation?

A. 3

B. 48

C. 93

D. 96

1.30 e223b909-79a8-46de-a366-14993592a4c7 · Easy

$$x = 8$$

$$y = x^2 + 8$$

The graphs of the equations in the given system of equations intersect at the point x, y in the xy-plane. What is the value of y?

A. 8

B. 24

C. 64

D. 72

1.31 e39fd227-fde8-40e7-97f3-74c7621ac276 · Easy

$$p + 34 = q + r$$

The given equation relates the variables p, q, and r. Which equation correctly expresses p in terms of q and r?

A. $p = q + r + 34$

B. $p = q + r - 34$

C. $p = -q - r + 34$

D. $p = -q - r - 34$

1.32 48cc6531-d8e0-4647-ac35-96cd60c2ea06 · Easy

$$x - 2 = 9$$

What is one possible solution to the given equation?

Student-produced response: _____

1.33 5dbe783c-201f-4b4a-86db-5f63b8c1f0cc · Easy

$$\frac{x^2}{25} = 36$$

What is a solution to the given equation?

A. 6

B. 30

C. 450

D. 900

1.34 79fcfcbb-499f-4f84-979c-55ad908c4878 · Easy

$$8x^2 - 40 = 32$$

What is the positive solution to the given equation?

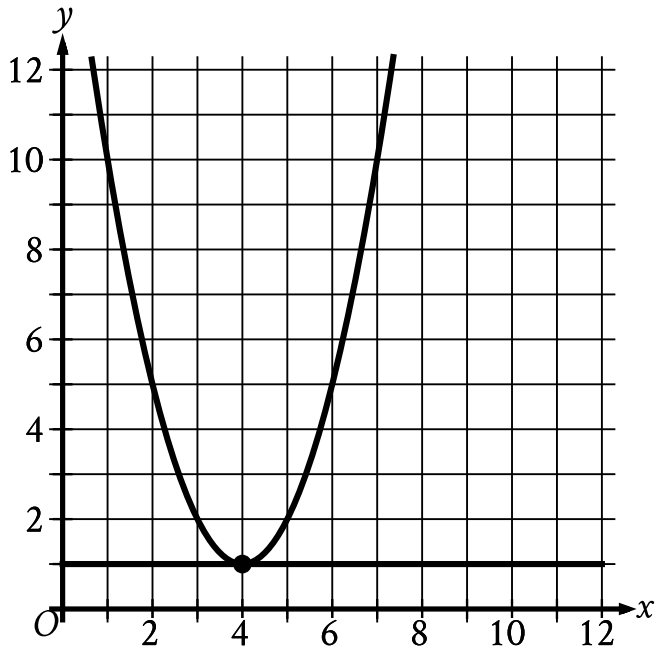
A. 3

B. 4

C. 9

D. 72

1.35 8900a10b-e0a1-45fb-9e6e-fa09017b372e · Easy



- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (4 comma 1).
  - The parabola passes through the following points:
    - (3 comma 2)
    - (4 comma 1)
    - (5 comma 2)
- For the line in the system:
  - The line is horizontal.
  - The line passes through the following points:
    - (0 comma 1)
    - (4 comma 1)

The graph of a system of a linear and a quadratic equation is shown. What is the solution x, y to this system?

A. $0, 0$

B. $-4, 1$

C. $4, -1$

D. $4, 1$

1.36 fe86204c-5062-494c-9970-e8162c2a17f3 · Easy

$$y - 57 = px$$

The given equation relates the positive numbers p, x, and y. Which equation correctly expresses y in terms of p and x?

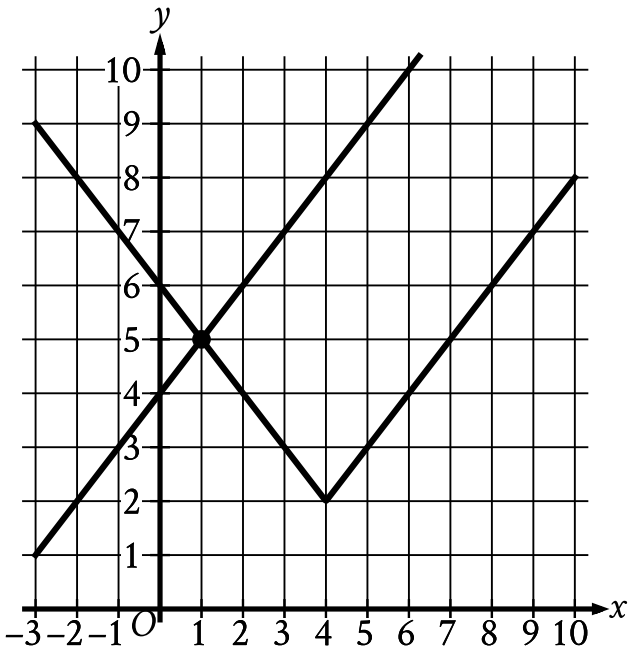
A. $y = 57x + p$

B. $y = px + 57$

C. $y = 57px$

D. $y = \frac{px}{57}$

1.37 7a8d2bc5-8edd-45eb-a918-924e8cba8608 · Easy



- For the absolute value function:
  - Moving from left to right:
    - The function slants sharply down to the point (4 comma 2).
    - The function then slants sharply up.
  - The function passes through the following points:
    - (1 comma 5)
    - (4 comma 2)
    - (6 comma 4)
- For the linear function:
  - The function slants sharply up from left to right.
  - The function passes through the following points:
    - (0 comma 4)
    - (1 comma 5)

The graph of a system of an absolute value function and a linear function is shown. What is the solution x, y to this system of two equations?

A. $-1, 5$

B. $0, 4$

C. $1, 5$

D. $4, 2$

1.38 a06c928d-bb2e-484a-9091-bfb1ee3246e4 · Easy

$$y = x^2 + 14x + 48$$

$$x + 8 = 11$$

The solution to the given system of equations is x, y. What is the value of y?

Student-produced response: _____

1.39 027912-DC · Easy

$x^2 = 64$

Which of the following values of x satisfies the given equation?

A. -8

B. 4

C. 32

D. 128

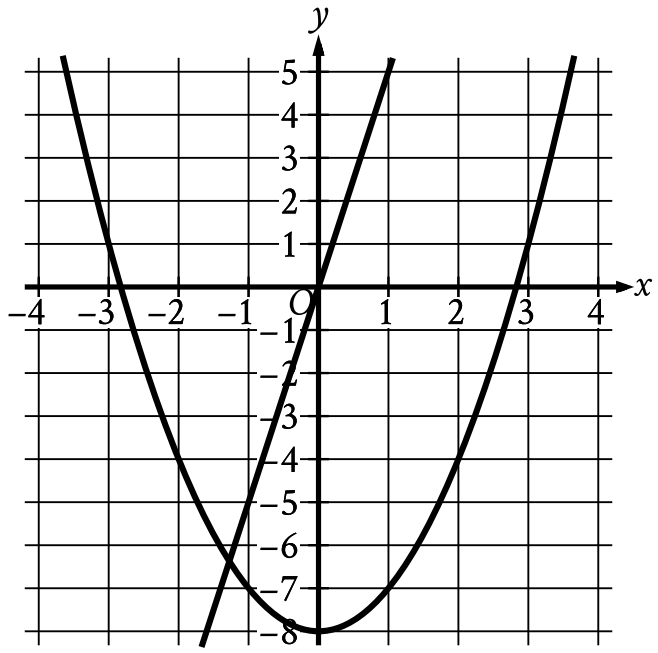
1.40 021044-DC · Easy

$\sqrt{x+4} = 11$

What value of x satisfies the equation above?

Student-produced response: _____

1.41 086aecb2-e874-4750-a452-88004556f2cf · Easy



- For the line in the system:
  - The line slants sharply up from left to right.
  - The line passes through the following points:
    - (negative 1 comma negative 5)
    - (0 comma 0)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at the point (0 comma negative 8).
  - The parabola passes through the following points:
    - (negative 1 comma negative 7)
    - (0 comma negative 8)
    - (1 comma negative 7)
- 1 intersection of the line and the parabola is shown in the graph.

The graph of a system of a linear and a quadratic equation is shown. Which system of equations is represented by the graph?

A. $y = -5x$ $y = x^2 + 8$

B. $y = -5x$ $y = x^2 - 8$

C. $y = 5x$ $y = x^2 + 8$

D. $y = 5x$ $y = x^2 - 8$

1.42 1b6bf314-ce44-468e-8236-7d3efe7c0c5f · Easy

$$x + 5 = 14$$

$$y = 4x^2 + 4$$

At what point x, y do the graphs of the equations in the given system intersect?

A. $9, 324$

B. $9, 328$

C. $14, 4$

D. $14, 788$

1.43 10aa6d6b-7baa-40a7-91f3-b2715a65ebc6 · Medium

$$x = 3$$

$$y = 15 - x^2$$

A solution to the given system of equations is x, y. What is the value of xy?

A. 432

B. 54

C. 45

D. 18

1.44 08143-DC · Medium

$y = x^2 - 1$
$y = 3$

When the equations above are graphed in the xy-plane, what are the coordinates (x, y) of the points of intersection of the two graphs?

A. $(2,3)$ and $(-2,3)$

B. $(2,4)$ and $(-2,4)$

C. $(3,8)$ and $(-3,8)$

D. $(\sqrt{2},3)$ and $(-\sqrt{2},3)$

1.45 04588-DC · Medium

$v^2 = \frac{LT}{m}$

The formula above expresses the square of the speed v of a wave moving along a string in terms of tension T, mass m, and length L of the string. What is T in terms of m, v, and L ?

A. $T = \frac{mv^2}{L}$

B. $T = \frac{m}{v^2 L}$

C. $T = \frac{mL}{v^2}$

D. $T = \frac{L}{mv^2}$

1.46 72653ce5-1c87-4942-ba7c-3afa5b87ce0e · Medium

$$38x^2 = 389$$

What is the negative solution to the given equation?

Student-produced response: _____

1.47 75080822-5747-4d72-a303-ce2f5788815a · Medium

$$d - 30d + 30 - 7 = -7$$

What is a solution to the given equation?

Student-produced response: _____

1.48 8c08419c-177e-4194-8d6c-68c4aa493232 · Medium

$$x^2 - 5x - 24 = 0$$

What is the sum of the solutions to the given equation?

Student-produced response: _____

1.49 3972a3e1-4431-4102-8724-89e23f92d34f · Medium

$$v = -\frac{w}{150x}$$

The given equation relates the distinct positive numbers v, w, and x. Which equation correctly expresses w in terms of v and x?

A. $w = -150vx$

B. $w = -\frac{150v}{x}$

C. $w = -\frac{x}{150v}$

D. $w = v + 150x$

1.50 ae0f3b56-b349-4863-90be-5dfd21f6eebd · Medium

$$\frac{-54}{w} = 6$$

What is the solution to the given equation?

Student-produced response: _____

1.51 0b84481f-8e4b-4584-b01c-7c57d2cb39f6 · Medium

$$\frac{x + 9x - 9}{x + 9} = 7$$

What is the solution to the given equation?

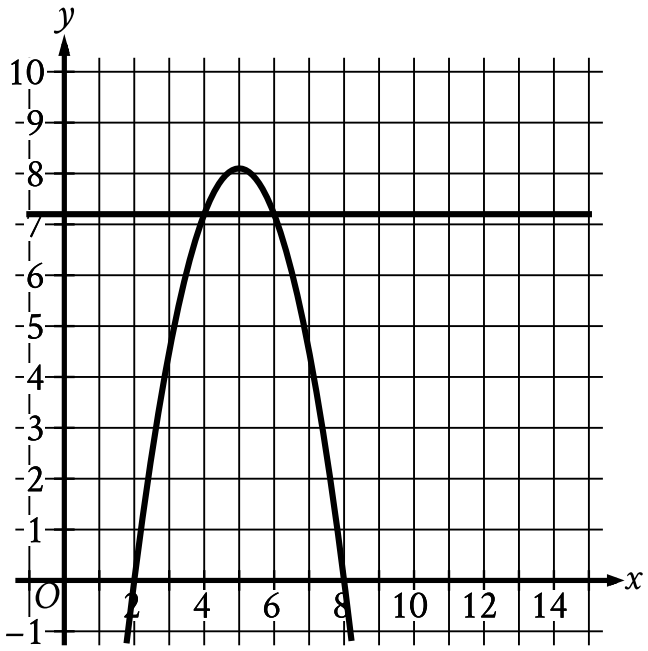
A. 7

B. 9

C. 16

D. 63

1.52 0b32b9eb-51a8-404e-b6ad-8166b494f610 · Medium



- For the line in the system:
  - The line is horizontal.
  - The line passes through the following points:
    - (0 comma 7.2)
    - (4 comma 7.2)
    - (6 comma 7.2)
- For the parabola in the system:
  - The parabola opens downward.
  - The vertex is at the point (5 comma 8.1).
  - The parabola passes through the following points:
    - (2 comma 0)
    - (4 comma 7.2)
    - (6 comma 7.2)
    - (8 comma 0)

The graph of a system of a linear equation and a quadratic equation is shown. A solution to the system is x, y. What is a possible value of x?

A. 4

B. 5

C. 7.2

D. 8

1.53 7b75e7d1-53d8-4028-84e1-be077af6c888 · Medium

$$b - 72 = \frac{x}{y}$$

The given equation relates the positive numbers b, x, and y. Which equation correctly expresses x in terms of b and y?

A. $x = \frac{b - 72}{y}$

B. $x = by - 72$

C. $x = \frac{by - 72}{y}$

D. $x = by - 72y$

1.54 133d99b4-1511-4e22-8582-1c56921faa65 · Medium

$$x - 16x - 10x + 7x + 17 = 0$$

What is a positive solution to the given equation?

Student-produced response: _____

1.55 cbd984f8-3214-4590-b53a-6195d99f9659 · Medium

$$x^2 = -841$$

How many distinct real solutions does the given equation have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

1.56 028796-DC · Medium

In the xy-plane, what is the y-coordinate of the point of intersection of the graphs of $y=(x-1)^2$ and $y=2x-3$?

Student-produced response: _____

1.57 025768-DC · Medium

$x^2-x-1=0$

What values satisfy the equation above?

A. $x=1$ and $x=2$

B. $x=-\frac{1}{2}$ and $x=\frac{3}{2}$

C. $x=\frac{1+\sqrt{5}}{2}$ and $x=\frac{1-\sqrt{5}}{2}$

D. $x=\frac{-1+\sqrt{5}}{2}$ and $x=\frac{-1-\sqrt{5}}{2}$

1.58 5041d89b-afb9-486c-83f0-f085fe47f197 · Medium

$$\frac{1}{7b}=\frac{11x}{y}$$

The given equation relates the positive numbers b, x, and y. Which equation correctly expresses x in terms of b and y?

A. $x=\frac{7by}{11}$

B. $x=y-77b$

C. $x=\frac{y}{77b}$

D. $x=77by$

1.59 3ba3f345-5069-44e0-84d8-8513595b8722 · Medium

If $\frac{42}{x} = 7x$, what is the value of $7x^2$?

A. 6

B. 7

C. 42

D. 294

1.60 b427042f-bfeb-436d-a1d2-2b4035d2e0dd · Medium

$$14j + 5k = m$$

The given equation relates the numbers j, k, and m. Which equation correctly expresses k in terms of j and m?

A. $k = \frac{m - 14j}{5}$

B. $k = \frac{1}{5}m - 14j$

C. $k = \frac{14j - m}{5}$

D. $k = 5m - 14j$

1.61 d92caabb-d558-4de0-984a-c30db0f20ddd · Medium

$$-4x^2 - 7x = -36$$

What is the positive solution to the given equation?

A. $\frac{7}{4}$

B. $\frac{9}{4}$

C. 4

D. 7

1.62 0de0c801-a394-401e-86d9-293cbb485a91 · Medium

$$p = 20 + \frac{16}{n}$$

The given equation relates the numbers p and n, where n is not equal to 0 and $p > 20$. Which equation correctly expresses n in terms of p?

A. $n = \frac{p - 20}{16}$

B. $n = \frac{p}{16} + 20$

C. $n = \frac{p}{16} - 20$

D. $n = \frac{16}{p - 20}$

1.63 bf79b193-f7b7-4f48-a581-1f6665ce3add · Medium

$$y = 4x$$

$$y = x^2 - 12$$

A solution to the given system of equations is x, y, where $x > 0$. What is the value of x?

Student-produced response: _____

1.64 3ebd45be-873f-4edf-8055-3fd8a34ced59 · Medium

$$P = N19 - C$$

The given equation relates the positive numbers P, N, and C. Which equation correctly expresses C in terms of P and N?

A. $C = \frac{19 + P}{N}$

B. $C = \frac{19 - P}{N}$

C. $C = 19 + \frac{P}{N}$

D. $C = 19 - \frac{P}{N}$

1.65 36a520ca-4944-44ee-bfd7-ed88c78436ef · Medium

$$5x^2 - 37x - 24 = 0$$

What is the positive solution to the given equation?

A. $\frac{3}{5}$

B. 3

C. 8

D. 37

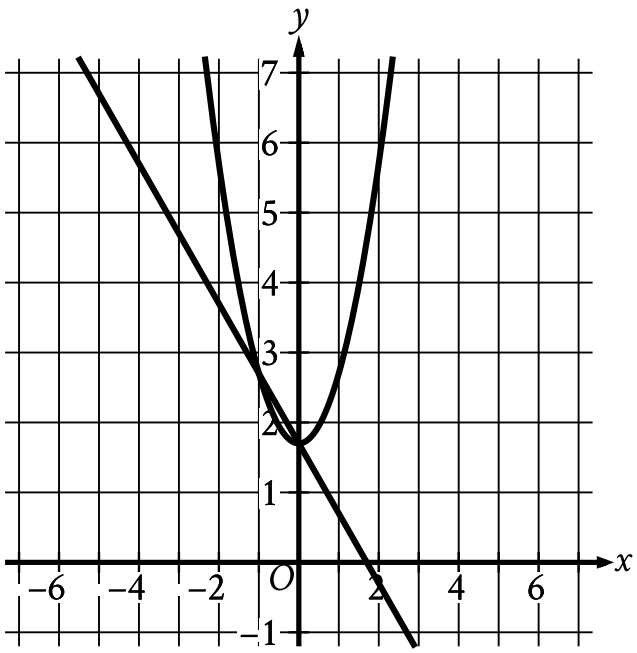
1.66 3fb87a51-838d-423b-8abf-0dca2a3cf712 · Medium

$$y = x^2 + 1.7$$

$$y = 1.7 - x$$

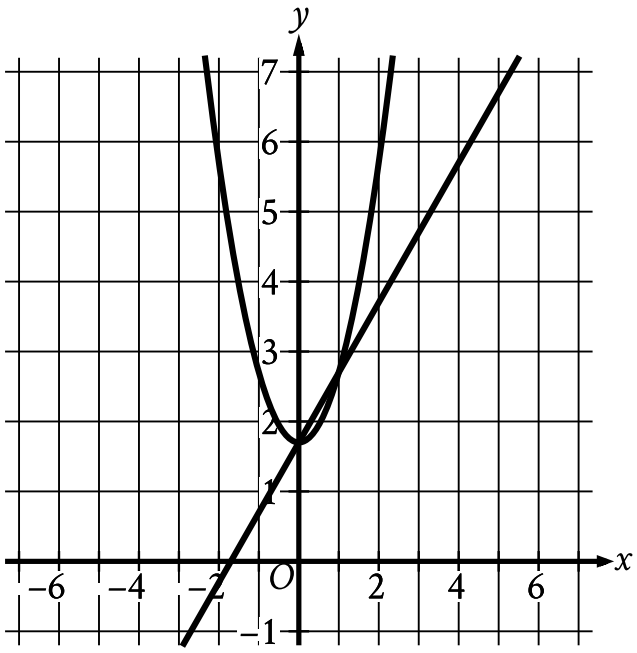
Which graph represents the given system of equations?

A.



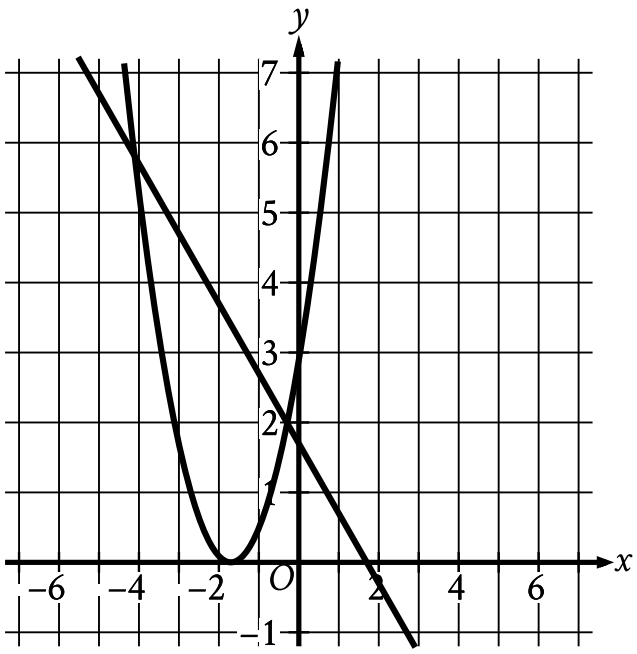
- For the line in the system:
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (0 comma 1.7).
  - The parabola passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
    - (1 comma 2.7)

B.



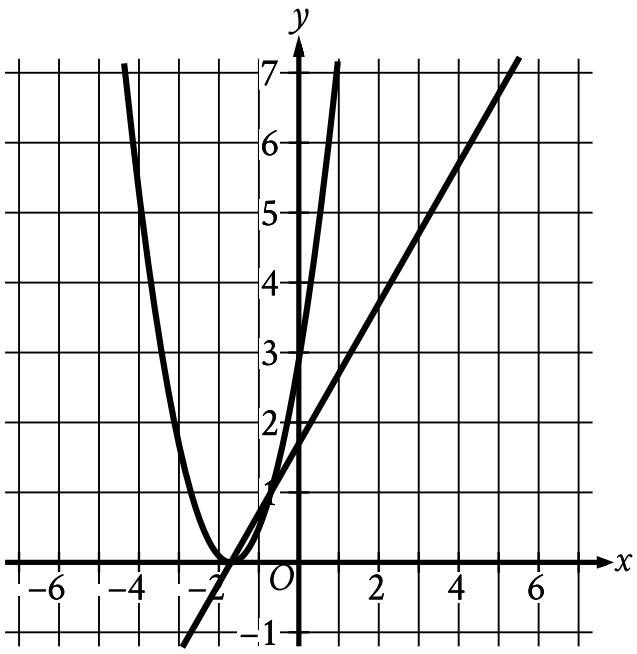
- For the line in the system:
  - The line slants sharply up from left to right.
  - The line passes through the following points:
    - (0 comma 1.7)
    - (1 comma 2.7)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (0 comma 1.7).
  - The parabola passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
    - (1 comma 2.7)

C.



- For the line in the system:
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (negative 1 comma 2.7)
    - (0 comma 1.7)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (negative 1.7 comma 0).
  - The parabola passes through the following points:
    - (negative 2.7 comma 1)
    - (negative 1.7 comma 0)
    - (negative 0.7 comma 1)

D.



- For the line in the system:
  - The line slants sharply up from left to right.
  - The line passes through the following points:
    - (0 comma 1.7)
    - (1 comma 2.7)
- For the parabola in the system:
  - The parabola opens upward.
  - The vertex is at point (negative 1.7 comma 0).
  - The parabola passes through the following points:
    - (negative 2.7 comma 1)
    - (negative 1.7 comma 0)
    - (negative 0.7 comma 1)

1.67 11250-DC · Medium

$x^2+x-12=0$

If a is a solution of the equation above and $a>0$, what is the value of a ?

Student-produced response: _____

1.68 135d0742-47f5-4297-a337-b8715edbb016 · Medium

$$p = \frac{k}{4j + 9}$$

The given equation relates the distinct positive numbers p, k, and j. Which equation correctly expresses $4j + 9$ in terms of p and k?

A. $4j + 9 = \frac{k}{p}$

B. $4j + 9 = kp$

C. $4j + 9 = k - p$

D. $4j + 9 = \frac{p}{k}$

1.69 070619-DC · Medium

$$x + y = 17$$
$$xy = 72$$

If one solution to the system of equations above is (x, y), what is one possible value of x ?

Student-produced response: _____

1.70 2aca4259-cade-40c9-b16c-5635ca1d9ac5 · Medium

If $4\sqrt{2x} = 16$, what is the value of $6x$?

A. 24

B. 48

C. 72

D. 96

1.71 46dc5769-2263-4a0c-adbe-bfe4c54562bc · Medium

$$z^2 + 10z - 24 = 0$$

What is one of the solutions to the given equation?

Student-produced response: _____

1.72 05314-DC · Medium

$y = x^2$
$2y + 6 = 2(x + 3)$

If (x, y) is a solution of the system of equations above and x > 0, what is the value of xy ?

A. 1

B. 2

C. 3

D. 9

1.73 035714-DC · Medium

$$x^2 = 6x + y$$
$$y = -6x + 36$$

A solution to the given system of equations is (x, y). Which of the following is a possible value of xy ?

A. 0

B. 6

C. 12

D. 36

1.74 015647-DC · Medium

$T = 0.01(P - 40{,}000)$

In a city, the property tax T, in dollars, is calculated using the formula above, where P is the value of the property, in dollars. Which of the following expresses the value of the property in terms of the property tax?

A. $P = 100T - 400$

B. $P = 100T + 400$

C. $P = 100T - 40{,}000$

D. $P = 100T + 40{,}000$

1.75 4f0d6e7c-87b5-4c6c-85c3-06cd2c4aedbe · Medium

$$5x + 42x - 5 = 0$$

Which of the following is a solution to the given equation?

A. $-\frac{5}{2}$

B. $-\frac{5}{4}$

C. $-\frac{4}{5}$

D. $-\frac{2}{5}$

1.76 b930ff78-e7c0-4bf0-a4c8-db513178d545 · Medium

$$x^2 + y + 10 = 10$$

$$8x + 16 - y = 0$$

The solution to the given system of equations is x, y. What is the value of x?

A. -16

B. -4

C. 2

D. 8

1.77 d431ba6d-965f-4df7-a8ee-357948a9d74e · Medium

Which quadratic equation has exactly one distinct real solution?

A. $x + 15^2 = 0$

B. $x + 15^2 = -45$

C. $x + 15^2 = 45$

D. $x + 15^2 = 135$

1.78 aeb387c2-21cb-4d5a-ab7d-88f31fa270af · Medium

$$x^2 = 2222$$

What is the positive solution to the given equation?

Student-produced response: _____

1.79 09153-DC · Medium

The speed of sound in dry air, v, can be modeled by the formula $v = 331.3 + 0.606T$, where T is the temperature in degrees Celsius and v is measured in meters per second. Which of the following correctly expresses T in terms of v ?

A. $T = \frac{v + 0.606}{331.3}$

B. $T = \frac{v - 0.606}{331.3}$

C. $T = \frac{v + 331.3}{0.606}$

D. $T = \frac{v - 331.3}{0.606}$

1.80 025718-DC · Medium

An oceanographer uses the equation $s = \frac{3}{2}p$ to model the speed s, in knots, of an ocean wave, where p represents the period of the wave, in seconds. Which of the following represents the period of the wave in terms of the speed of the wave?

A. $p = \frac{2}{3}s$

B. $p = \frac{3}{2}s$

C. $p = \frac{2}{3} + s$

D. $p = \frac{3}{2} + s$

1.81 027749-DC · Medium

$(x+2)(x+3)=(x-2)(x-3)+10$

Which of the following is a solution to the given equation?

A. 1

B. 0

C. -2

D. -5

1.82 ae003425-4a05-45a2-b10f-6a51a4a26073 · Medium

$$y = x - 2x + 4$$

$$y = 6x - 12$$

Which ordered pair x, y is the solution to the given system of equations?

A. $0, 2$

B. $-4, 2$

C. $2, 0$

D. $2, -4$
